A pavement is a hard crust placed on soil for the purpose of placed on the soil for the purpose of providing smooth and strong surface on which vehicals can move.

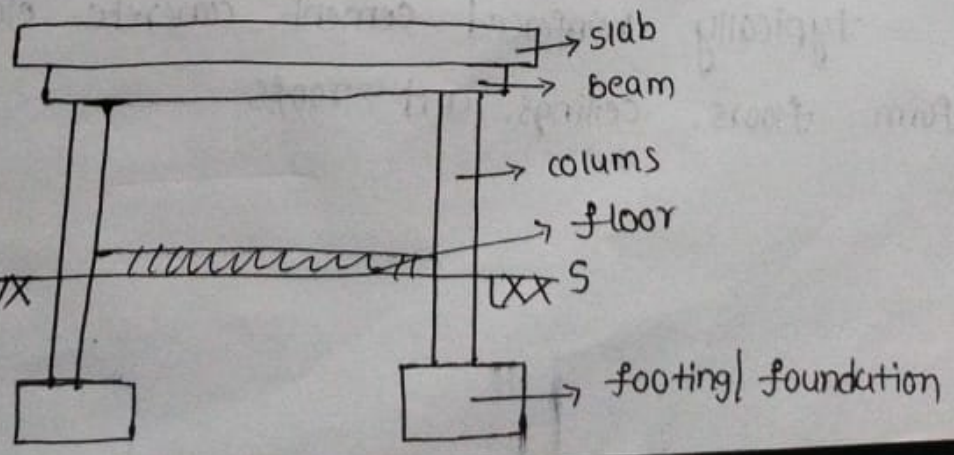
* Structural Engineering :-

Structural Engineering is the branch of civil Engineering which deals with planning, design, construction and maintainence of structures such as buildings, bridges and dams

Scope of Structural Engineering :-

1. Detailing and specification of structure
2. Design the buildings to resist all kinds of loads
3. structural engineering deals with structural members and Materials
4. project supervision and monitary
5. Investigation on collapse structures
6. Maintainence of structures
7. Repair and Rehabilation of the structure

Structure & Basic structural Engineering



Structure is the assemblage of two or more basic structural elements such as beam, foundation, column, slab and walls etc..

The basic structural elements are

1. Footing / Foundation
2. column
3. Beam
4. slab
5. Floor
6. wall
7. stairs

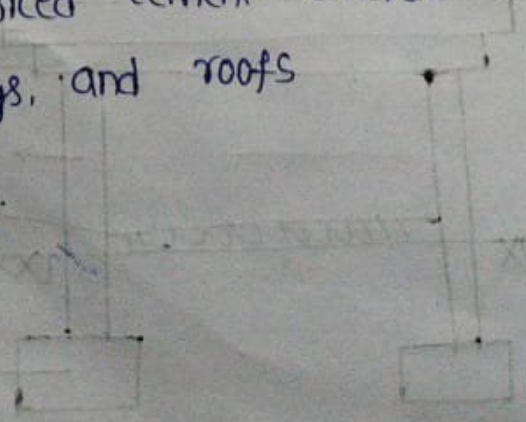
Footing :- Foundation is the lower most structure in any building, it will transfer the load from superstructure to the soil.

Column :- It is the vertical member of a structure it is made up with reinforced cement concrete (RCC) the columns are transferred the load fro beam to footing

Beam :- It is a horizontal component of a building structure it is made of (RCC)

* It will transfer the load from slab to column

slab :- slab is defined as a flat, horizontal and typically reinforced cement concrete element us to form floors, ceilings, and roofs



Floor :- A floor in building construction is a levelled surface which can support the objects and occupanties etc.

Wall :- A wall is a structure that defines an area, carries a load provides the security, shelter, sound proofing and decorative

Stairs :- stairs are structural elements design to bridge a large vertical distance between lower and higher levels by dividing it into smaller vertical distances

Engineering :-

bricks should be low.

Types of Aggregates :-

* Types of aggregates based on size
* Types of aggregates based on shape

Types of Aggregates According to size :

⇒ **Coarse Aggregates :-**

* The aggregate particle which retains on 4.75mm sieve is known as coarse aggregate
* It occupies 45% to 60% volume of total aggregates.

⇒ **Fine Aggregates :-**

* Aggregate which passed through 4.75mm sieve known as fine aggregate
* It occupies 35% to 40% volume of total aggregates

* Classification of Aggregates Based of shape :

⇒ Rounded Aggregates :-

* Rounded aggregates are totally nature - made shape
* It is available in the form of sea shore gravel
* It has voids (32-33%), hence offers lesser workability
* It has bond strength slightly good compared to rounded aggregate
* It is not suitable for high - strength concrete.

⇒ Angular Aggregate :-

* Angular Aggregate has well defined edges.
* These are obtained by crushing the rocks.
* It has voids (38 - 45%) hence offer less workability.
* It gives high specific area
* It has good bond strength between the ingredients in concrete

⇒ Flaky Aggregates :-

* Flaky aggregate thickness is less compared to length and width
* Thickness is less than the 60% of its mean dimension
* It is not suitable for concrete making.

⇒ Elongated Aggregate :-

* Elongated aggregate length is larger to compare to width and thickness.
* Length of aggregate is greater than 180% of its mean dimension.
* It is not suitable for concrete making

⇒ Flaky and Elongated Aggregates :-

* If length of aggregate is larger than its width and with is larger than its thickness then it is said to be flaky and elongated aggregates.
* It is not suitable for concrete making.

Geotechnical Engineering :-

Geotechnical Engineering is the branch of civil Engineering which deals with analysis, design, and construction of foundations, slopes, retaining walls and embankments, canals, tunnels and other systems that are made of or supported by soil.

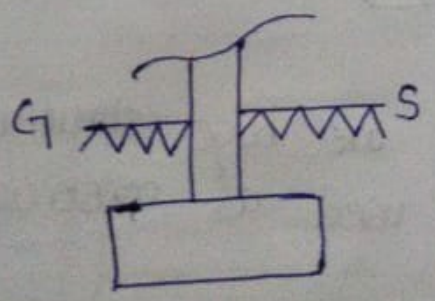
Scope of geotechnical Engineering :-

1. Evolute the soil conditions and earth materials
2. To determine the stability of ground, include slope and soil deposits.
3. To determine the safe bearing capacity of the soil.
4. To predict the type of foundation required for the particular type of building
5. To potentially analyse the hazars like land slides, earthquakes other seismic activities etc....
6. To study the techniques involved in improvement of soil with less bearing capacity.

Applications of GTE :-

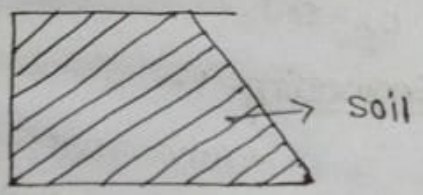
⇒ Geotechnical Engineering has vast applications in the construction of various civil engineering works.

* **Foundation**:- Every civil Engineering structure whether it is a building a bridge or a dam is founded on are below the surface of earth



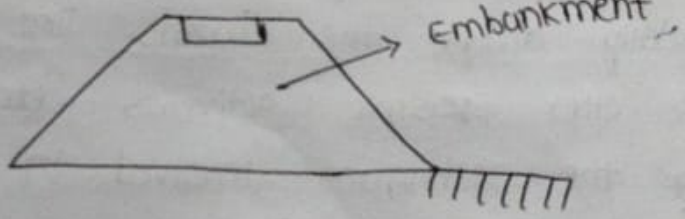
Retaining structures :-

When sufficient space is not available for a mass of soil to spread and form a safe slope, a structure is required to retained a soil the structure is known as retaining structure



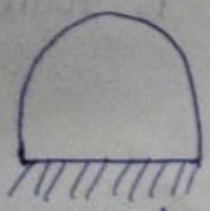
Stability of slopes :-

If the soil surface is not horizontal there is a component of weight of the soil which tens to move downward direction and thus causes instability of slope



Underground Structures :-

The design and construction of underground structu such as tunnels and conduit, required evolution of sources existed by the soil on the structures



tunnel

Earth Dam :- Earth Dams are huge structures in which soil is used as construction material.

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slope,
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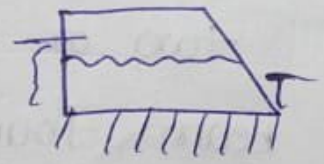

payment Design :-

A pavement is a hard crust placed on soil for the purpose of placed on the soil for the purpose of providing smooth and strong surface on which vehicals can move.

* Various disciplines in civil Engineering

* Geotechnical Engineering
* structural Engineering
* Transportation Engineering
* water resources Engineering
* Enviornmental Engineering

2. chimney

Types of Bricks

1. Based on strength (BIS)

The bricks are classified as per buro of Indian standards (BIS) as follows.

S. NO	Class	compressive strength N/mm²
1.	Class 35	35
2.	Class 30	30
3.	Class 25	25
4.	Class 20	20
5.	Class 17	17
6.	Class 15	15
7.	Class 12	12
8.	Class 10	10
9.	Class 7.5	7.5

Based on quality :-

Bricks are classified based on the quality are as foll

1. First class
2. second class
3. Third class
4. Fourth class
5. special types of Bricks

No	Class of Bricks	First class	second class	Third class	fourth class	special type of Bricks
1.	compressive strength (N/mm^2)	10 to 20	7	<7	<3.5	40
2.	water absorption for 24 hrs (%)	15 to 20	18 to 20	20 to 25	more than 30%	5%
3.	colour	Deep red	copper colour	Red yellow	Reddish Black	Deep red
4.	sound	Metallic ring	Metallic ring	Dull	Dull	Metallic ring
5.	Texture	uniform Texture	uniform	uniform	non-uniform	uniform
6.	surface	smooth rectangular	smooth rectangular	smooth rectangular	unsmooth	smooth rectangular
7.	size	190x90x90 mm	190x90x90 mm	190x90x90mm	190x90x90mm	190x90x90mm
8.	Moulding	Table	Ground Moulding	Ground moulding	Ground mouldling	Table mouldling
9.	Burning	klins (or) chimney	klins (or) chimney	klins	klins	klins
10.	use	Brick walls, RCC Brick work	unimportant masonry work	Temporary masonry structures	Filling of Ground area for floor	Important structures like Dams, and bridges

* Characteristics of Bricks :-

* The bricks should be uniform in colour, size and shape
* standard size of brick should be maintained that is max 190 x 90 x 90 mm
* The standard size of frog should be maintained that is 100 x 40 x 10 mm
* Bricks should be sound and compact
* Bricks should be free from cracks, air bubbles and stones with sharp and square edges.
* Bricks should not observe more than 15 to 20% of water when immersed in water 24 hours.
* weight of 1 clay brick is 3kg
* unit weight of clay brick should be 1600 to 1900 kg/m^3
* the compressive strength of bricks should be in the range of 7.5 to 35 N/mm^2
* The percentage of soliable salts should be ~~in the~~ not a exceed 25% in burnt brick
* Bricks should not change in volume when wetted
* Brick should neither over burn nor under burn
* The thermal conductivity of bricks should be low.